

Susan Liang

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Final-Year Ph.D. Student, Computer Science

Looking for Full-Time Opportunities in 2026

EDUCATION

University of Rochester

NY, USA

Ph.D. of Computer Science

Sept. 2022 – Dec. 2026 (Expected)

- Advisor: Prof. Chenliang Xu
- GPA: 4/4

University of Chinese Academy of Sciences

Beijing, China

Bachelor of Computer Science

Sept. 2018 – Jul. 2022

- Advisor: Prof. Shiguang Shan
- GPA: 3.90/4, Rank: 4/104

RESEARCH INTERESTS

- Vision-Language Model & Agent
- Audio-Video Generation
- Audio-Visual World Modeling

PUBLICATION

VidHarness: Evolving Agent Harnesses for Cost-Efficient Long Video Understanding

Susan Liang, Jianmin Wu, Daxiang Dong

arXiv preprint, 2026

AdaTurn: Budget-Aware Test-Time Scaling for Active Visual Perception Agents

Susan Liang, Chao Huang, Filippos Bellos, Jing Bi, Jason J Corso, Chenliang Xu

arXiv preprint, 2026

Omni-Judge: Can Omni-LLMs Serve as Human-Aligned Judges for Text-Conditioned Audio-Video Generation?

Susan Liang, Chao Huang, Filippos Bellos, Yolo Yunlong Tang, Qianxiang Shen, Jing Bi, Luchuan Song, Zeliang Zhang, Jason Corso, Chenliang Xu

The IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2026

π -AVAS: Can Physics-Integrated Audio-Visual Modeling Boost Neural Acoustic Synthesis?

Susan Liang, Chao Huang, Yunlong Tang, Zeliang Zhang, Chenliang Xu

International Conference on Computer Vision (ICCV), 2025

BinauralFlow: A Causal and Streamable Approach for High-Quality Binaural Speech Synthesis with Flow Matching Models

Susan Liang, Dejan Markovic, Israel D. Gebru, Steven Krenn, Todd Keebler, Jacob Sandakly, Frank Yu, Samuel Hassel, Chenliang Xu, Alexander Richard

The International Conference on Machine Learning (ICML), 2025

Rethinking Audio-Visual Adversarial Vulnerability from Temporal and Modality Perspectives

Zeliang Zhang*, **Susan Liang***, Daiki Shimada, Chenliang Xu

The Thirteenth International Conference on Learning Representations (ICLR), 2025

High-Quality Visually-Guided Sound Separation from Diverse Categories

Chao Huang, **Susan Liang**, Yapeng Tian, Anurag Kumar, Chenliang Xu

17th Asian Conference on Computer Vision (ACCV), 2024 (**Best Paper Honorable Mention**)

Language-Guided Joint Audio-Visual Editing Via One-Shot Adaptation

Susan Liang, Chao Huang, Yapeng Tian, Anurag Kumar, Chenliang Xu

17th Asian Conference on Computer Vision (ACCV), 2024

AV-NeRF: Learning Neural Fields for Real-World Audio-Visual Scene Synthesis

Susan Liang, Chao Huang, Yapeng Tian, Anurag Kumar, Chenliang Xu

Conference on Neural Information Processing Systems (NeurIPS), 2023

Neural Acoustic Context Field: Rendering Realistic Room Impulse Response With Neural Fields

Susan Liang, Chao Huang, Yapeng Tian, Anurag Kumar, Chenliang Xu

International Conference on Computer Vision Workshops (ICCVW), 2023

UniCon: Unified Context Network for Robust Active Speaker Detection

Yuanhang Zhang*, Susan Liang*, Shuang Yang, Xiao Liu, Zhongqin Wu, Shiguang Shan, Xilin Chen

ACM International Conference on Multimedia (ACM MM), 2021 (Oral)

Suppression Is Not Forgetting: Residual Recoverability in Visual Concept Unlearning for VLMs

Zhangyun Tan, Zeliang Zhang, Jiani Liu, Susan Liang, Yolo Y. Tang, Lisha Chen, Chenliang Xu

arXiv preprint, 2026

BVB: Benchmarking Agentic Video Understanding via Programmatic Reconstruction in Blender

Yolo Y. Tang, Daiki Shimada, Jiayue Meng, Jing Bi, Pinxin Liu, Yicheng Wang, Yunzhong Xiao, Zhangyun Tan, Zeliang Zhang, Chao Huang, Susan Liang, Qianxiang Shen, Luchuan Song, Ali Vosoughi, Mingqian Feng, Melika

Filvantorkaman, Chenliang Xu

arXiv preprint, 2026

SurgAtlas: A Large-Scale Surgical Video-Language Dataset with 2,391 Hours of Open and Minimally Invasive Surgery

Filippos Bellos, Andre S. Gala-Garza, Miaowei Wang, Alyssa M. Hardin, Ahmad M. Hider, Li Yayuan, Jing Bi,

Susan Liang, Chenliang Xu, Donald S. Likosky, Jason J. Corso

arXiv preprint, 2026

VERIFY: A Benchmark of Visual Explanation and Reasoning for Investigating Multimodal Reasoning Fidelity

Jing Bi, Junjia Guo, Guangyu Sun, Susan Liang, Luchuan Song, Yunlong Tang, Jinxi He, Jiarui Wu, Ali Vosoughi, Chen Chen, Jason J. Corso, Chenliang Xu

The Third Annual Conference on Language Modeling (CoLM), 2026

Asynchronous Temporal Modeling with Two-Agent Framework for Streaming Dense Video Captioning

Yolo Yunlong Tang, Chao Huang, Susan Liang, Jing Bi, Yicheng Wang, Daiki Shimada, Chenliang Xu

The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

Bridging Facial Understanding and Animation via Language Models

Luchuan Song, Pinxin Liu, Haiyang Liu, Zhenchao Jin, Yolo Yunlong Tang, Zichong Xu, Susan Liang, Jing Bi, Jason J. Corso, Chenliang Xu

The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

When to Think and When to Look: Uncertainty-Guided Lookback

Jing Bi, Filippos Bellos, JunJia Guo, Yayuan Li, Chao Huang, Yolo Yunlong Tang, Luchuan Song, Susan Liang, Zhongfei Zhang, Jason J. Corso, Chenliang Xu

The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

Caption Anything in Video: Fine-grained Object-centric Captioning via Spatiotemporal Multimodal Prompting

Yunlong Tang, Jing Bi, Chao Huang, Susan Liang, Daiki Shimada, Hang Hua, Yunzhong Xiao, Yizhi Song, Pinxin Liu, Mingqian Feng, Junjia Guo, Zhuo Liu, Luchuan Song, Ali Vosoughi, Jinxi He, Liu He, Zeliang Zhang, Jiebo Luo, Chenliang Xu

The Association for the Advancement of Artificial Intelligence (AAAI), 2026 (Best Demonstration Award Runner-Up)

Why Reasoning Matters? A Survey of Advancements in Multimodal Reasoning

Jing Bi, Susan Liang, Xiaofei Zhou, Pinxin Liu, Junjia Guo, Yunlong Tang, Luchuan Song, Chao Huang, Ali Vosoughi, Guangyu Sun, Jinxi He, Jiarui Wu, Shu Yang, Daoan Zhang, Chen Chen, Lianggong Bruce Wen, Zhang Liu, Jiebo Luo, Chenliang Xu

arXiv preprint, 2025

High-Quality Sound Separation Across Diverse Categories via Visually-Guided Generative Modeling

Chao Huang, **Susan Liang**, Yapeng Tian, Anurag Kumar, Chenliang Xu

International Journal of Computer Vision (IJCV), 2025

ZeroSep: Separate Anything in Audio with Zero Training

Chao Huang, Yuesheng Ma, Junxuan Huang, **Susan Liang**, Yunlong Tang, Jing Bi, Wenqiang Liu, Nima Mesgarani, Chenliang Xu

The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025

Harnessing the Computation Redundancy in ViTs to Boost Adversarial Transferability

Jiani Liu*, Zhiyuan Wang*, Zeliang Zhang*, Chao Huang, **Susan Liang**, Yunlong Tang, Chenliang Xu

The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025

MMPerspective: Do MLLMs Understand Perspective? A Comprehensive Benchmark for Perspective Perception, Reasoning, and Robustness

Yolo Y. Tang, Pinxin Liu, Zhangyun Tan, Mingqian Feng, Rui Mao, Chao Huang, Jing Bi, Yunzhong Xiao,

Susan Liang, Hang Hua, Ali Vosoughi, Luchuan Song, Zeliang Zhang, Chenliang Xu

The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025

Video-LMM Post-Training: A Deep Dive into Video Reasoning with Large Multimodal Models

Yolo Y. Tang, Jing Bi, Pinxin Liu, Zhenyu Pan, Zhangyun Tan, Qianxiang Shen, Jiani Liu, Hang Hua, Junjia Guo,

Yunzhong Xiao, Chao Huang, Zhiyuan Wang, **Susan Liang**, Xinyi Liu, Yizhi Song, Junhua Huang, Jia-Xing Zhong,

Bozheng Li, Daiqing Qi, Ziyun Zeng, Ali Vosoughi, Luchuan Song, Zeliang Zhang, Daiki Shimada, Han Liu, Jiebo Luo,

Chenliang Xu

arXiv preprint, 2025

VIDCOMPOSITION: Can MLLMs Analyze Compositions in Compiled Videos?

Yunlong Tang, Junjia Guo, Hang Hua, **Susan Liang**, Mingqian Feng, Xinyang Li, Rui Mao, Chao Huang, Jing Bi,

Zeliang Zhang, Pooyan Fazli, Chenliang Xu

The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025

Generative AI for Cel-Animation: A Survey

Yunlong Tang, Junjia Guo, Pinxin Liu, Zhiyuan Wang, Hang Hua, Jia-Xing Zhong, Yunzhong Xiao, Chao Huang,

Luchuan Song, **Susan Liang**, Yizhi Song, Liu He, Jing Bi, Mingqian Feng, Xinyang Li, Zeliang Zhang, Chenliang Xu

arXiv preprint, 2025

Will the Inclusion of Generated Data Amplify Bias Across Generations in Future Image Classification Models?

Zeliang Zhang, Xin Liang, Mingqian Feng, **Susan Liang**, Chenliang Xu

arXiv preprint, 2024

Scaling Concept With Text-Guided Diffusion Models

Chao Huang, **Susan Liang**, Yunlong Tang, Yapeng Tian, Anurag Kumar, Chenliang Xu

arXiv preprint, 2024

Learning to Transform Dynamically for Better Adversarial Transferability

Rongyi Zhu*, Zeliang Zhang*, **Susan Liang**, Zhuo Liu, Chenliang Xu

Conference on Computer Vision and Pattern Recognition (CVPR), 2024

Random Smooth-based Certified Defense against Text Adversarial Attack

Zeliang Zhang, Wei Yao, **Susan Liang**, Chenliang Xu

Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2024

Video Understanding with Large Language Models: A Survey

Yunlong Tang*, Jing Bi*, Siting Xu*, Luchuan Song, **Susan Liang**, Teng Wang, Daoan Zhang, Jie An, Jingyang Lin,

Rongyi Zhu, Ali Vosoughi, Chao Huang, Zeliang Zhang, Feng Zheng, Jianguo Zhang, Ping Luo, Jiebo Luo, Chenliang Xu

IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)

UNICON+: ICTCAS-UCAS Submission to the AVA-ActiveSpeaker Task at ActivityNet Challenge 2022

Yuanhang Zhang*, **Susan Liang***, Shuang Yang, Xiao Liu, Zhongqin Wu, Shiguang Shan

IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2022

ICTCAS-UCAS-TAL Submission to the AVA-ActiveSpeaker Task at ActivityNet Challenge 2021

Yuanhang Zhang*, **Susan Liang***, Shuang Yang, Xiao Liu, Zhongqin Wu, Shiguang Shan
IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2021

* indicates equal contribution.

RESEARCH EXPERIENCE

Streaming Audio-Video Generation Models

May 2026 – Aug. 2026

*GenAI Team, Global Monetization Product & Technology, **ByteDance***

Advisor: Dr. Chuchu Han

- Proposed a training-free autoregressive audio-video generation framework without causal post-training.
- Designed a KV cache management strategy to reduce memory footprint and preserve long-term consistency.

Vision-Language Agents for Long-Video Understanding

May 2025 – Aug. 2025

*Coding Agent Team, AI Cloud, **Baidu***

Advisor: Daxiang Dong

- Proposed an agent evolution framework to design video harnesses for cost-efficient long-video understanding.
- Introduced an MCTS-based harness evolution strategy with multi-fidelity validation for efficient evolution.
- Designed a mixture-of-harnesses mechanism to improve the generalization of evolved harnesses.

Position-Conditioned Binaural Audio Generation

May 2024 – Aug. 2024

*Audio Team, Reality Labs Research, **Meta***

Advisors: Dr. Dejan Markovic and Dr. Alexander Richard

- Designed a flow matching model to generate high-quality position-conditioned binaural audio.
- Proposed a causal, streaming U-Net architecture for real-time inference.

Low-Level Image Semantic Correspondence

Sept. 2021 – Mar. 2022

*Vision and Learning Lab, **University of California, Merced***

Advisors: Prof. Ming-Hsuan Yang and Dr. Taihong Xiao

- Proposed a self-supervised low-level content learning approach for fine-grained semantic correspondence.
- Exploited contrastive learning and cycle consistency to learn discriminative and consistent features.

Vector Graphics Learning and Generation

Jun. 2021 – Aug. 2021

*Institute for AI Industry Research, **Tsinghua University***

Advisors: Dr. Yizhi Wang and Dr. Hao Xu

- Developed an encoder-decoder model to convert raster images to vector graphics.
- Designed a differentiable rasterization pipeline to enable gradient-based optimization of vector graphics.

Active Speaker Detection with Joint Audio-Video Understanding

Oct. 2020 – Apr. 2021

*VIPL, Institute of Computing Technology, **Chinese Academy of Sciences***

Advisors: Prof. Shiguang Shan and Dr. Shuang Yang

- Developed an audio-visual fusion scheme to detect when each visible person in the video is speaking.
- Proposed a permutation-equivariant layer that processes all speakers in the scene simultaneously.
- Exploited skew-symmetry of inter-speaker relations to improve interpretability and reduce memory and FLOPs.

Face Deformation Field Generation and Lip Reading

Feb. 2020 – Sept. 2020

*VIPL, Institute of Computing Technology, **Chinese Academy of Sciences***

Advisors: Prof. Shiguang Shan and Dr. Shuang Yang

- Developed an encoder-decoder model to reconstruct facial deformation fields (optical flow).
- Designed a self-supervised training approach for deformation fields that requires no annotations.
- Combined deformation fields with grayscale face images to improve lip-reading performance.

AWARDS

AAAI 2026 Best Demonstration Award Runner-Up	Jan. 2026
ICML 2025 Scholar Award	Jul. 2025
ACCV 2024 Best Paper Honorable Mention	Dec. 2024
UCAS Outstanding Thesis Award	Jun. 2022
ActivityNet CVPR 2022 Workshop AVA Active Speaker Detection Challenge First Place	Jun. 2022
ActivityNet CVPR 2021 Workshop AVA Active Speaker Detection Challenge First Place	Jun. 2021
UCAS Overseas Graduate Studies Fellowship	Aug. 2021
UCAS Academy Fellowship	Sept. 2019 - Jun. 2021

ACADEMIC SERVE

Organizer: ICCV 2026 Trustworthy FMs Workshop

Reviewer: ICLR 2027, NeurIPS 2026, CVPR 2026, WACV 2026, NeurIPS 2025, ICCV 2025, ICML 2025, CVPR 2025, ICLR 2025, AAAI 2025, NeurIPS 2024, AAAI 2024, CVPR 2024